SATHYABAMA
INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

SecureMailScope: AI-Assisted Cryptographic Security Posture Assessment for Secure Email Communications



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- Team Name :BRIX
- Theme:BLOCKCHAIN AND CYBER SECURITY
- PS Category- Software
- Team ID- DEX26251

SECURE MAILSCOPE

PROPOSED SOLUTION

Secure MailScope is an AI-assisted tool that reads PCAP captures and flags the crypto weaknesses visible in SMTP, IMAP and POP3 traffic.

HOW IT ADDRESSES THE PROBLEM

- Finds which email protocols are in the capture and where TLS is used
- Checks the certificates and other crypto details it can see
- Rules set how serious each issue is, with advice on how to fix it
- Puts everything into dashboards and reports in one place

INNOVATION & UNIQUENESS

- Ties packet-capture analysis to security checks you can actually explain
- Turns low-level packet details into findings ranked by priority
- Leaves room to add ML-based anomaly detection later

SECURE MAILSCOPE WORKFLOW



RESEARCH

Wireshark Docs | RFC 5321 | RFC 9051 | RFC 1939

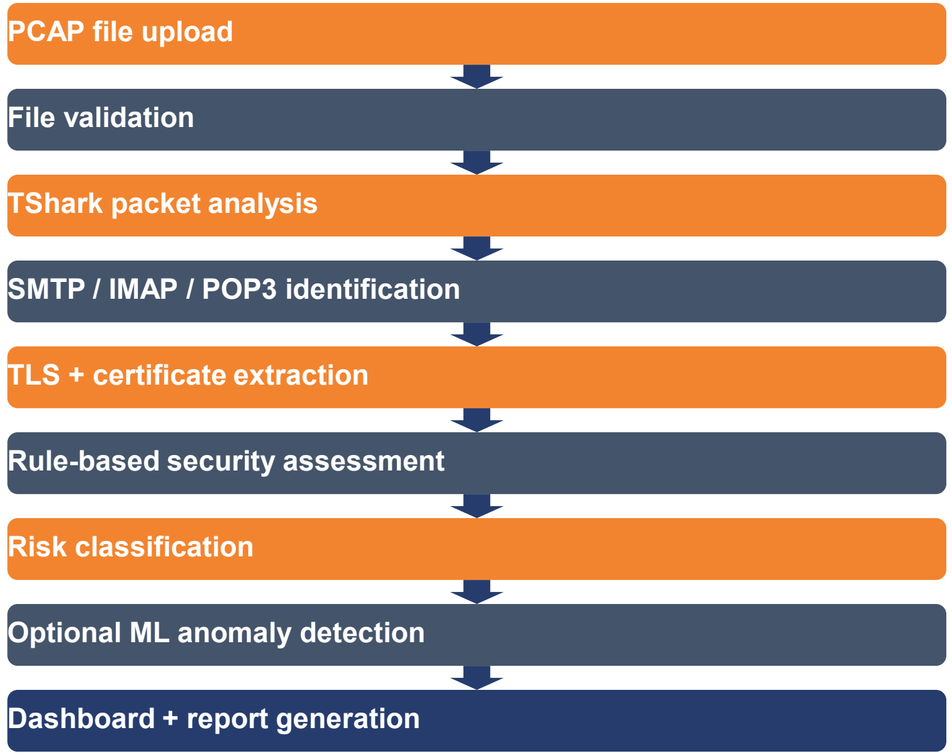
TECHNICAL APPROACH

TECHNOLOGIES TO BE USED

- Python for the backend and security analysis
- FastAPI for the backend API
- TShark to break down packets and spot protocols
- cryptography to read certificate details
- React + TypeScript for the web interface
- scikit-learn for optional ML anomaly detection
- Supabase for optional login and scan history
- HTML, PDF and JSON for reports

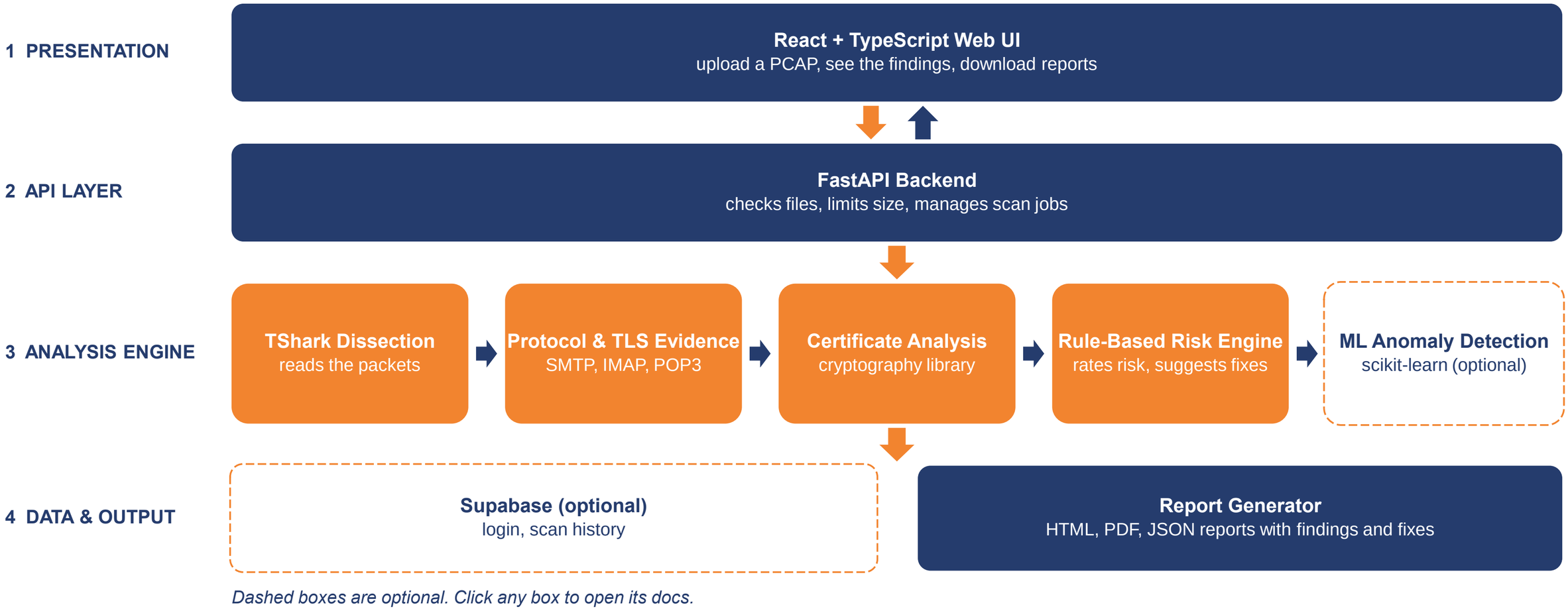
In short: the PCAP is checked and accepted, then we pull out the protocol and TLS details we can see, run them through our security checks, rank the findings by risk and suggest fixes.

IMPLEMENTATION ARCHITECTURE



RESEARCH: TShark Manual | cryptography | FastAPI | React | scikit-learn | Supabase

SYSTEM ARCHITECTURE



Dashed boxes are optional. Click any box to open its docs.

RESEARCH FastAPI | TShark Manual | Supabase | scikit-learn

FEASIBILITY AND VIABILITY

FEASIBILITY

- Built on proven open-source tools and Python libraries
- Software only, so no special hardware is needed
- A rule-based engine is enough for the first version
- Easy to demo with suitable test PCAP files

CHALLENGES & RISKS

- Incomplete captures can hide the TLS handshake
- Encrypted traffic limits what we can see at the application level
- Mail servers are all set up a little differently
- The ML part could give false alarms
- Large PCAP files take more processing

MITIGATION STRATEGIES

- Validate files and parse them in isolation
- Clearly flag when the evidence is incomplete
- Document every rule and finding
- Test with controlled captures
- Trust ML results only after evaluating them
- Limit file size and resource use

FEASIBILITY STATEMENT

A focused prototype is realistic with the tools that already exist. More protocols, deeper detection and validated ML can be added step by step.

RESEARCH

TShark Manual | scikit-learn | RFC 8446

BUSINESS CANVAS

MISSION & VISION

Mission: Make email cryptography checks easier to access, understand and act on.

Vision: Help organizations spot email security weak points before something goes wrong.

MARKET & COMPETITIVE ANALYSIS

Users: SMBs, IT and security teams, consultants, teaching labs.

Alternatives: packet-analysis tools, TLS checkers and enterprise email security platforms.

What sets us apart: PCAP inspection, explained findings, severity and fix suggestions in one tool.

BUSINESS PERSPECTIVE

Where it could go later:
- A free community and educational edition
- Paid assessment and reporting features
- Deployment and support for organizations
- Authorized security assessment services

FINANCIAL PLANNING & REQUIREMENTS

What we need to start:
- A development laptop and environment
- Open-source tools and test PCAPs
- Hosting or compute, only if we deploy online
- Time for testing, documentation and upkeep

Cost plan: start with a small open-source prototype.

RESEARCH RFC 8314 NIST SP 800-52r2 Wireshark Docs

RESEARCH AND REFERENCES

OFFICIAL TECHNICAL REFERENCES

1. **Wireshark documentation**
   wireshark.org/docs
2. **TShark manual**
   wireshark.org/docs/man-pages/tshark.html
3. **Python cryptography**
   cryptography.io/en/latest
4. **FastAPI**
   fastapi.tiangolo.com
5. **scikit-learn**
   scikit-learn.org/stable
6. **RFC 3207**
   rfc-editor.org/rfc/rfc3207
7. **RFC 8446 (TLS 1.3)**
   rfc-editor.org/rfc/rfc8446

GITHUB REPOSITORY LINKS

Our project — Secure MailScope
https://github.com/pitu2k/Securemailscope

testssl.sh — TLS / STARTTLS checker
https://github.com/testssl/testssl.sh

SSLyze — TLS/SSL scanner
https://github.com/nabla-c0d3/sslyze

Zeek — network analysis framework
https://github.com/zeek/zeek